

Hongye Liu

hongye.liu@duke.edu | +919-564-5878

EDUCATION

Duke University	PhD Candidate	09/2024 – 03/2027 (Expected)
– Adviser: Ricardo Henao – Research Interest: NLP, RL, RLHF, RLVR, Agent, Model Evaluation, Control Generation		
KAUST	PhD Student	09/2023 – 09/2024
– Adviser: Ricardo Henao – Project: Learning to Substitute Words with Model-based Score Ranking		
Imperial College London	MSc in Artificial Intelligence	10/2021 – 10/2022
– Graduated with High Merit – Group project: Drug discovery with Graph Networks (Supervised by: Paul Bilokon) – Individual project: Emotion Recognition in Speech and further Modalities (Supervised by: Bjoern Schuller) – Courses: Computational Optimisation: 85.1/100; Deep Learning: 78.36/100		
Beijing Institute of Technology, Zhuhai	BSc in Applied Statistics	09/2016 – 06/2020
– GPA: 3.74/5 (Ranking: 1% - Equal to First Class Honors) – Courses: Python: 91/100; Bayesian Statistics: 97/100; Non-Parametric Statistics: 97/100; Machine Learning: 96/100		

PUBLICATIONS

Publications

- Hongye Liu, Rongmei Lin et al., Customer-Agent: Overcoming Context Limitations in Ultra-Long Shopping Trajectories via Tool-Augmented Agents and RLVR (Under Review)
- Dhanajit Brahma, Hongye Liu, Ricardo Henao A Probabilistic Framework for Multimodal Open-Set Test-Time Adaptation
- Hongye Liu, Liang Ding, Ricardo Henao, Learning to Control Summaries with Score Ranking, ACL-Finding 2026
- Hongye Liu, Dhanajit Brahma, Ricardo Henao, Calibrating Evaluation Metrics for QA and Summarization. ACL-Finding 26
- Hongye Liu, Ricardo Henao, Learning to Substitute Words with Model-based Score Ranking. NAACL-Main 2025
- Mingzhou Xu, Longyue Wang, Derek F. Wong, Hongye Liu, Linfeng Song, Lidia S. Chao, Shuming Shi, and Zhaopeng Tu. A Benchmark for Zero Pronoun Recovery and Translation. EMNLP-Main 2022.
- Yang Shen, Hongye Liu et al., Intelligent Flaw Detection of X-ray Images Based on Deep Learning. (Springer Science and Business Media B.V., 2021), vol. 105, pp. 558–566.

Patents

- A Method and Device of Bilingual Data for Web Fiction Translation via Self-Training (Hong Kong patent) 2021
- A Method and Device for Automatic Alignment and Automatic Translation of Web Fiction (Mainland patent) 2021
- A Method and Device of Large-scale Data Filtering for Machine Translation (Mainland patent) 2021
- A Ray Defect Detection Method based on Mask R-CNN model (Mainland patent) 2018

INTERNSHIPS

Amazon	Applied Scientist Intern	Palo alto	05/2026 – 08/2026
Manager: Xi Chen			
– Compress memory as Embedding via model alignment. – Utilizing memory for downstream task personalized question answering.			
Amazon	Applied Scientist Intern	Seattle	09/2025 – 12/2025
Manager: Rongmei Lin/ Besnik Fetahu			
– Introduced a long-context shopping QA benchmark (32k/64k), revealing bottlenecks of LLMs on long shopping trajectories. – Proposed a Customer Agent Framework with verifiable SFT data synthesis and tool-based reasoning over externalized shopping context files, enabling effective multi-turn tool use and long-context reasoning. – Demonstrated strong performance and generalization across internal and external reasoning benchmarks.			
Tencent AI Lab, NLP Group	Research & Engineering Intern	Shenzhen	09/2020 – 09/2021
Supervisor: Longyue Wang/ Zhaopeng Tu			
Webnovel Translation			

- Extracted CH-EN Webnovel data and optimized corpus alignment algorithm to reduce the translation omission issue.
- Built domain corpus selections, back-translation and NMT model training pipeline.
- Developed front-end web and launch the translation engine (Link: longyuewang.com/nmt).

Large-scale Parallel Corpus Filtering

- Integrated N-gram and LM to filter high-quality corpus and used Moore-lewis to select out related field data.

Zero pronoun Resolution

- Built test sets for zero pronoun translation from alignment, ZP labels and Part-of-Speech in four different domains.
- Assisted statistical data distribution, experiment and error analysis.

Beijing Genomics Institute (BGI)

AI Algorithm Intern

Shenzhen

09/2019 – 01/2020

Supervisor: Daoling Huang

NLP-AI Medical

- Designed a data pipeline based on electronic medical records to connect the upstream and downstream of the hospital.
- Built word segmentation, NER, relationship extraction to knowledge graph and knowledge query algorithm pipeline.

Biological Gene-AI platform

- Investigated related papers and reproduced their work.
- Utilized Docker to package the models as mirror images and deployed them on the platform.

Integrated Design Project

- Build AutoEncoder, Resnet, and DeepFm models with Pytorch for feature extraction.
- Carry out cross-omics academic research based on real biological genetic data.

Search Engine for Scientific Research

- Crawled biomedical articles including gene data on the Internet.
- Deployed Lucene engine and built indexes to queried related articles.

OTHER PROJECTS

Emotion Recognition in Speech and further Modalities

Master Essay

02/2022 – 10/2022

- Pre-trained an audio model and visual model with extensively-used features.
- Proposed an audio-visual model comprised of Wav2vec in audio and Vision Transformer in visual parts.
- Reached the SOTA performance of the 3rd ABAW challenge.

Drug Discovery with Graph Neural Networks

Master Group Project

02/2022 – 04/2022

- Predict the water solubility and toxicity of a molecule with the Graph Neural Network.
- Developed a front-end web with Streamlit.
- Built an automatic DevOps pipeline with Docker and GitLab CI/CD.

Application of Deep Learning on X-ray testing

Workshop

05/2019 – 05/2020

- Collected the industrial data, designed AI algorithm for X-ray testing detection and achieved 81.3% recognition rate.
- Presented an academic report in the 2nd Guangdong-Hong Kong- Macao Greater Bay Area NDT forum.

Application of GAN in Seismic Image Recognition of Salt Mine

Workshop

03/2019 – 05/2019

- Implemented image segmentation using the GAN model with Pytorch to recognize salt mine seismic images.
- Won the national second prize and presented an academic report at Sun Yat-sen University.

AWARDS

- Meritorious Winner at Mathematical Contest in Modeling (MCM) in 2018
- 2nd Prize at The First China Geoscience Big Data Mining and AI Challenge in 2019
- 3rd Prize at The MathorCup University Mathematical Modeling Challenge in 2018
- Awarded Outstanding Student and Outstanding League Member (3 times) since 2016

ACTIVITIES

- Vice President at the School Mathematical Modeling Association 09/2017 – 07/2018
- Assistant at the University Student Affairs Center 09/2016 – 11/2017
- Volunteer at the Guangdong QuanAi Social Service Center 07/2018 – 09/2018
- Volunteer for KAUST Commencement 2023, Jeddah, Saudi Arabia 12/2023